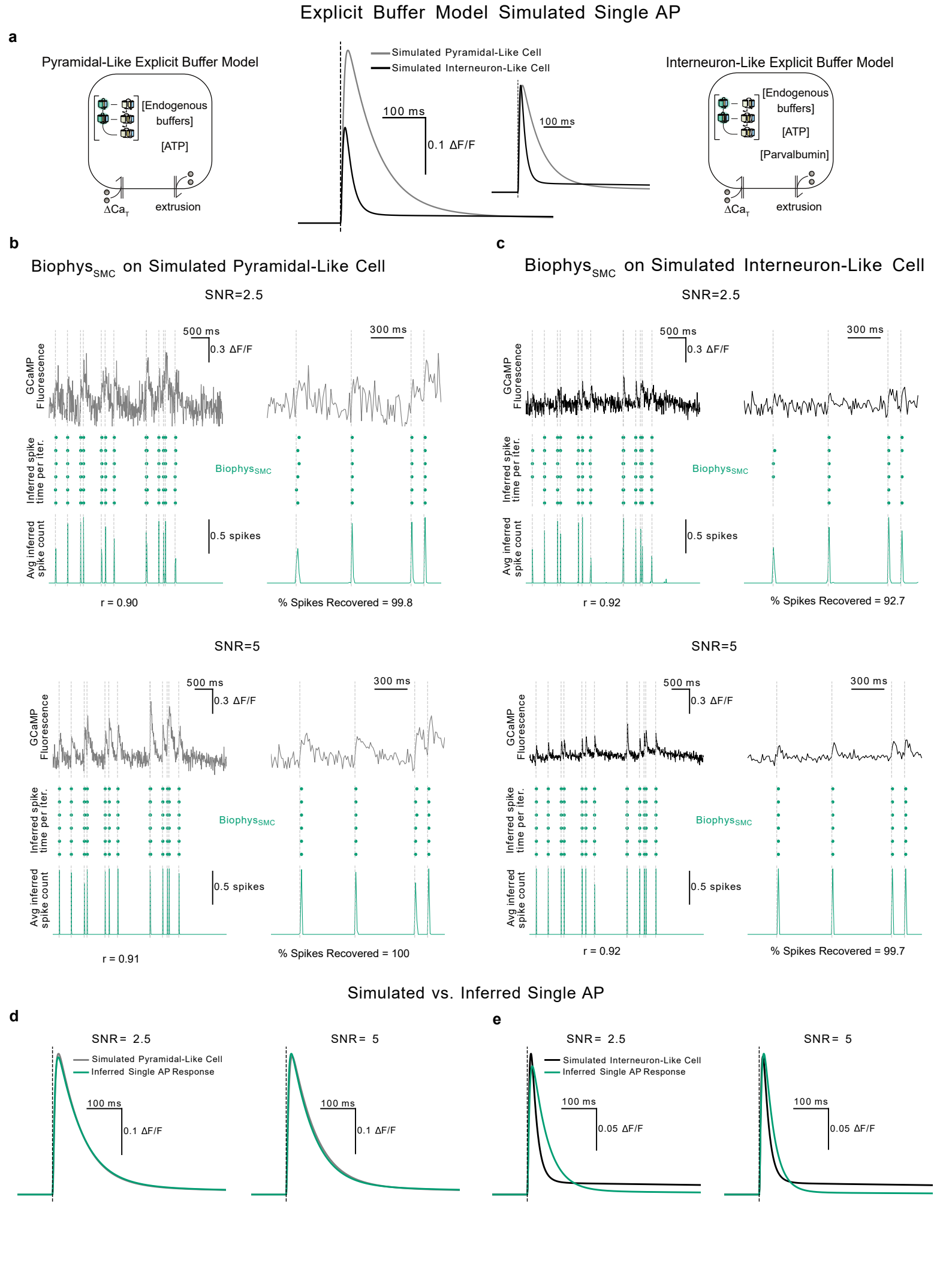

Explicit Buffer Model Simulated Single AP
a
Pyramidal-Like Explicit Buffer Model
[Endogenous buffers]
[ATP]
ΔCa_T
extrusion
Simulated Pyramidal-Like Cell
Simulated Interneuron-Like Cell
100 ms
0.1 ΔF/F
100 ms
Interneuron-Like Explicit Buffer Model
[Endogenous buffers]
[ATP]
[Parvalbumin]
ΔCa_T
extrusion
b
Biophys_SMC on Simulated Pyramidal-Like Cell
SNR=2.5
500 ms
0.3 ΔF/F
300 ms
GCaMP Fluorescence
Inferred spike time per iter.
Biophys_SMC
Avg inferred spike count
0.5 spikes
r = 0.90
% Spikes Recovered = 99.8
c
Biophys_SMC on Simulated Interneuron-Like Cell
SNR=2.5
500 ms
0.3 ΔF/F
300 ms
GCaMP Fluorescence
Inferred spike time per iter.
Biophys_SMC
Avg inferred spike count
0.5 spikes
r = 0.92
% Spikes Recovered = 92.7
SNR=5
500 ms
0.3 ΔF/F
300 ms
GCaMP Fluorescence
Inferred spike time per iter.
Biophys_SMC
Avg inferred spike count
0.5 spikes
r = 0.91
% Spikes Recovered = 100
SNR=5
500 ms
0.3 ΔF/F
300 ms
GCaMP Fluorescence
Inferred spike time per iter.
Biophys_SMC
Avg inferred spike count
0.5 spikes
r = 0.92
% Spikes Recovered = 99.7
Simulated vs. Inferred Single AP
d
SNR= 2.5
Simulated Pyramidal-Like Cell
Inferred Single AP Response
100 ms
0.1 ΔF/F
SNR= 5
100 ms
0.1 ΔF/F
e
SNR= 2.5
Simulated Interneuron-Like Cell
Inferred Single AP Response
100 ms
0.05 ΔF/F
SNR= 5
100 ms
0.05 ΔF/F